Grade 12 — Life Sciences Lebanese curriculum	**Practice Exam 2** Chapters 1 & 2 — Energy · Linear Momentum	Duration: 2 hours Total: 20 points

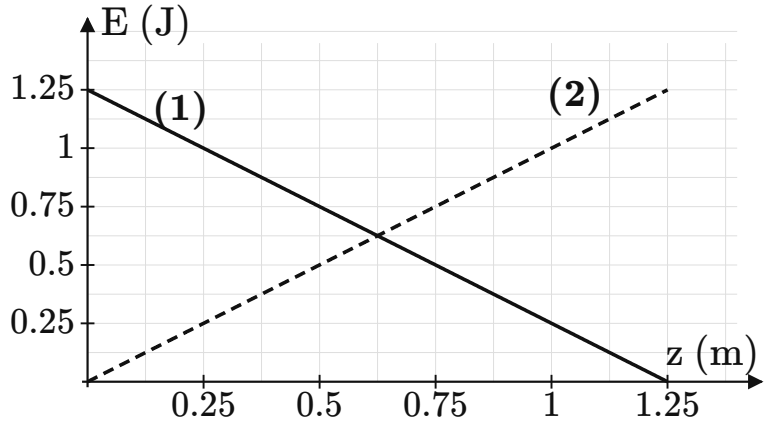
This exam is formed of three obligatory exercises. The use of a non-programmable calculator is allowed.

Take, in all exercises: $g = 10\ \text{m/s}^2$.

Exercise 1 — A ball thrown upwards (7 points)

A ball (B), considered as a particle of mass $m = 100$ g, is thrown vertically upwards from a point O with a speed v_0. An appropriate device gives, during the ascent, the kinetic energy KE of (B) and the gravitational potential energy GPE of the system [(B), Earth] as functions of the altitude z of (B) above O (Doc. 1). Air resistance is neglected in Parts 1 and 2.

> The horizontal plane through O is the reference level of gravitational potential energy.



Doc. 1

Part 1 — Reading the graph

1.1 Identify, with justification, the curve that represents KE and the one that represents GPE.

1.2 Using Doc. 1, calculate the speed v_0 and give the maximum altitude h reached by (B).

Part 2 — The mechanical energy

2.1 Calculate the mechanical energy of the system [(B), Earth] at O and at the highest point. Conclude.

2.2 Calculate the speed of (B) when it passes through the altitude $z = 0.8$ m.

Part 3 — With air resistance

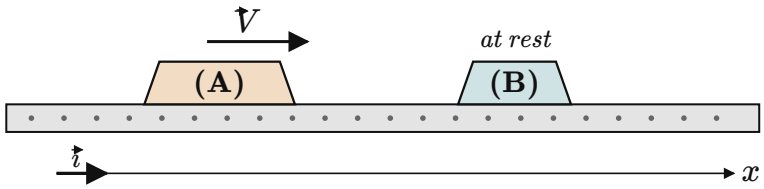
In reality, (B), thrown with the same speed v_0, rises only to the altitude $h' = 1$ m. The air resistance is a force $\vec{f}$ of constant magnitude, opposite to the motion.

3.1 Calculate the variation of the mechanical energy of the system [(B), Earth] between O and the new highest point. Deduce the magnitude f.

3.2 Deduce the variation ΔU of the internal energy of the system [(B), air, Earth] during the ascent.

Exercise 2 — Collision on an air track (7 points)

On a horizontal air track, a glider (A) of mass $m_A = 400$ g, moving with the velocity $\vec{V}_A = 0.6\,\vec{i}$ (m/s), collides head-on with a glider (B) of mass $m_B = 200$ g, initially at rest (Doc. 2). Friction is neglected.



Doc. 2

Part 1 — Before the collision

1.1 Show that the system [(A), (B)] is pseudo-isolated.

1.2 Calculate the linear momentum $\vec{P}$ of the system before the collision.

Part 2 — After the collision

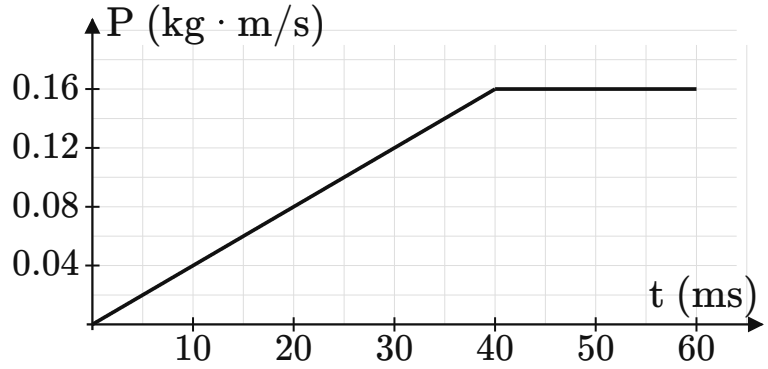
Just after the collision, (B) moves with the velocity $\vec{V}'_B = 0.8\,\vec{i}$ (m/s).

2.1 Determine the velocity $\vec{V}'_A$ of (A) just after the collision.

2.2 Show that the collision is elastic.

Part 3 — The force during the collision

Doc. 3 shows the magnitude P of the linear momentum of (B) during the collision, which lasts 40 ms.



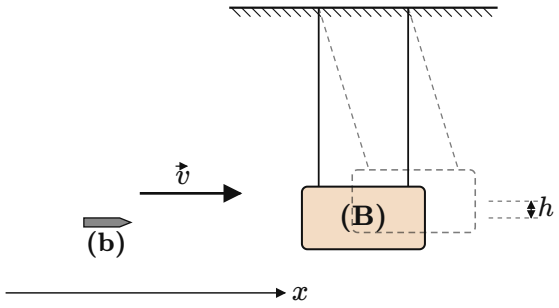
Doc. 3

3.1 Using Doc. 3, calculate the force $\vec{F}_{A/B}$ exerted by (A) on (B) during the collision, supposed constant.

3.2 Deduce the force $\vec{F}_{B/A}$ exerted by (B) on (A). Name the law used.

Exercise 3 — Measuring the speed of a bullet (6 points)

To measure the speed of a bullet (b) of mass $m = 10$ g, it is fired horizontally into a wooden block (B) of mass $M = 1.99$ kg, hanging at rest from light strings (Doc. 4). The bullet, moving with the velocity $\vec{v} = v\,\vec{i}$ where $v = 400$ m/s, gets embedded in the block, and the block rises. Air resistance is neglected.



Doc. 4

Part 1 — The impact

1.1 Calculate the linear momentum of the bullet just before the impact.

1.2 The system [(b), (B)] is pseudo-isolated during the impact. Determine the speed V of the block (with the bullet inside) just after the impact.

Part 2 — Energy

The horizontal plane through the initial position of the centre of mass of (B) is the reference level of gravitational potential energy.

2.1 Calculate the kinetic energy of the system [(b), (B)] just before and just after the impact. Deduce the variation of the internal energy of the system.

2.2 Using the conservation of the mechanical energy of the system [(b) + (B), Earth], determine the height h by which the block rises.

— End of the exam —